

3. Name the organic product: (a) but-2-ene + H_2 (Ni) (b) but-2-ene + Br_2 (c) chloroethane + OH^- (aq) (d) propan-1-ol heated under reflux with acidified dichromate (e) butan-2-ol warmed with acidified dichromate

C. Tests and selectivity

4. Describe a chemical test, with the observations, to tell apart: (a) hexane and hex-1-ene (b) propan-1-ol and 2-methylpropan-2-ol.

5. Propan-1-ol can be oxidized to propanal or to propanoic acid. How do the conditions decide which product forms?

6. Propene reacts with HBr . (a) Name the two possible products. (b) Which is the major product, and why?

D. Calculations

7. Ethanol is made industrially by $\text{CH}_2=\text{CH}_2 + \text{H}_2\text{O} \longrightarrow \text{CH}_3\text{CH}_2\text{OH}$. In one run, 14.0 kg of ethene gives 21.2 kg of ethanol. Calculate the theoretical yield and the percent yield.

8. 9.21 g of ethanol is completely oxidized to ethanoic acid. What mass of ethanoic acid forms?

E. Challenge

9. Starting only from ethene, water and the reagents in the data box, plan a synthesis of ethyl ethanoate. Give each step, its reagents and conditions, and the type of reaction.